JAMES RICHARDSON

London, United Kingdom

james.richardson@gmail.com

+44 7911 284756

LinkedIn: linkedin.com/in/james-richardson-ml-engineer

PROFESSIONAL SUMMARY

Machine Learning Engineer with 6 years of experience building and deploying production ML systems across fintech and healthcare sectors. I've led teams through end-to-end model development, from initial experimentation to monitoring in production. I'm particularly interested in solving real-world problems where model performance directly impacts business outcomes, and I've spent the last few years helping organisations move beyond proof-of-concepts into sustainable ML infrastructure. Currently based in London and open to hybrid opportunities.

WORK EXPERIENCE

Senior Machine Learning Engineer

Synapse Data Ltd

London, United Kingdom

www.synapsedata.io

January 2022 - Present

Architected and deployed a real-time fraud detection system processing 50,000+ transactions daily using gradient boosting, reducing false positives by 34% and saving the business GBP 1.2M annually

Established ML model governance framework including versioning, validation protocols, and rollback procedures, ensuring compliance with FCA regulations for financial services

Led a team of 2 junior ML engineers on feature engineering initiatives, designing 40+ new predictors from raw transaction data that improved model AUROC from 0.82 to 0.91

Built comprehensive monitoring dashboards tracking model drift, data quality metrics, and prediction distribution shifts, implementing automated alerts for performance degradation

Machine Learning Engineer

JPMorgan Chase

London, United Kingdom

www.jpmorganchase.com

November 2019 - December 2021

Developed end-to-end credit risk models using XGBoost and logistic regression, deployed across 5 business units and evaluated on 2M+ customer records quarterly

Optimised model inference pipeline reducing prediction latency from 800ms to 120ms through algorithmic improvements and code-level optimisations in Python

Collaborated with data engineering teams to build feature stores and ETL pipelines using PySpark on AWS, enabling daily model retraining and feature consistency

Conducted A/B testing on model variants in production, designing experiments that validated statistical significance of performance improvements before full rollout

Data Scientist

Deloitte Consulting

London, United Kingdom

www.deloitte.com

July 2018 - October 2019

Built classification and regression models for client engagements across retail, manufacturing, and public sector using scikit-learn and TensorFlow

Delivered predictive analytics solutions including demand forecasting for supply chain optimisation, achieving 18% reduction in inventory holding costs for a FTSE 250 client

Created automated data pipeline and reporting framework in Python that reduced manual analysis time from 20 hours to 2 hours per week

Presented model findings and recommendations to senior stakeholders, translating technical results into business impact narratives and implementation roadmaps

EDUCATION

MSc Machine Learning

University College London (UCL)

London, United Kingdom

Graduated: September 2018

Distinction

BSc Computer Science

University of Manchester

Manchester, United Kingdom

Graduated: June 2016

First Class Honours

TECHNICAL SKILLS

Languages and Frameworks: Python, SQL, R, Scala, PySpark, TensorFlow, PyTorch, scikit-learn, XGBoost, LightGBM

Platforms and Tools: AWS (SageMaker, S3, EC2, Lambda), Google Cloud Platform (BigQuery, Vertex AI), Jupyter, Git, Docker, Kubernetes, Terraform, ArgoCD

Machine Learning: Supervised learning, ensemble methods, time series analysis, feature engineering, model evaluation, hyperparameter tuning, A/B testing

Data Engineering: ETL pipelines, data warehousing, Apache Spark, data quality frameworks, feature stores

CERTIFICATIONS

AWS Certified Machine Learning - Specialty (2021)

Google Cloud Professional Data Engineer (2020)